JEE (ADVANCED) 2019 PAPER 2
PART-II CHEMISTRY

SECTION 1 (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : −1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get +4 marks;
  choosing ONLY (A) and (B) will get +2 marks;
  choosing ONLY (A) and (D) will get +2 marks;
  choosing ONLY (B) and (D) will get +2 marks;
  choosing ONLY (A) will get +1 mark;
  choosing ONLY (B) will get +1 mark;
  choosing ONLY (D) will get +1 mark;
  choosing no option (i.e. the question is unanswered) will get 0 marks; and
  choosing any other combination of options will get −1 mark.

Q.1 The cyanide process of gold extraction involves leaching out gold from its ore with CN^- in the presence of **Q** in water to form **R**. Subsequently, **R** is treated with **T** to obtain Au and **Z**. Choose the correct option(s)

(A) **Q** is O_2

(B) **T** is Zn

(C) **Z** is $[Zn(CN)_4]^{2-}$

(D) **R** is $[Au(CN)_4]^-$

Q.2 With reference to *aqua regia*, choose the correct option(s)

(A) *Aqua regia* is prepared by mixing conc. HCl and conc. HNO_3 in 3:1 (*v*/*v*) ratio

(B) Reaction of gold with *aqua regia* produces an anion having Au in +3 oxidation state

(C) Reaction of gold with *aqua regia* produces NO_2 in the absence of air

(D) The yellow colour of *aqua regia* is due to the presence of NOCl and Cl_2

Q.3 Consider the following reactions (unbalanced)

Zn + hot conc. $H_2SO_4 \rightarrow$ **G** + **R** + **X**
Zn + conc. $NaOH \rightarrow$ **T** + **Q**
G + H_2S + $NH_4OH \rightarrow$ **Z** (a precipitate) + **X** + **Y**

Choose the correct option(s)

(A) **Z** is dirty white in colour

(B) The oxidation state of Zn in **T** is +1

(C) **R** is a V-shaped molecule

(D) Bond order of **Q** is 1 in its ground state

Q.4 The ground state energy of hydrogen atom is −13.6 eV. Consider an electronic state Ψ of He^+ whose energy, azimuthal quantum number and magnetic quantum number are −3.4 eV, 2 and 0, respectively. Which of the following statement(s) is(are) true for the state Ψ?

(A) It is a 4*d* state

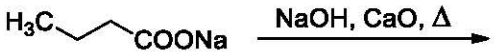
(B) It has 2 angular nodes

(C) It has 3 radial nodes

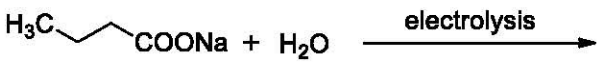
(D) The nuclear charge experienced by the electron in this state is less than 2*e*, where *e* is the magnitude of the electronic charge

Q.5 Which of the following reactions produce(s) propane as a major product?

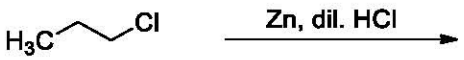
(A)



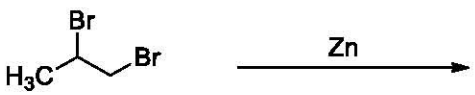
(B)



(C)

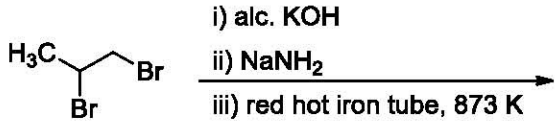


(D)

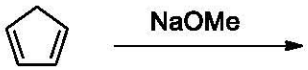


Q.6 Choose the correct option(s) that give(s) an aromatic compound as the major product

(A)



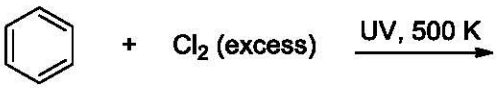
(B)



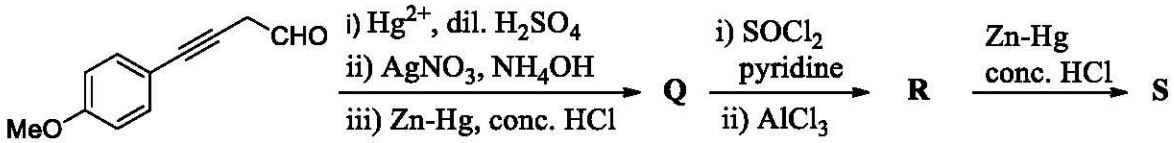
(C)



(D)

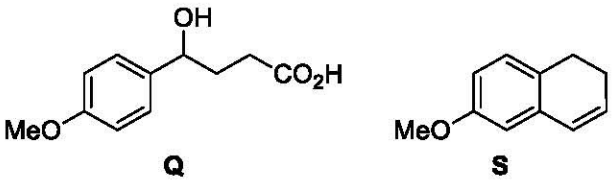


Q.7 Choose the correct option(s) for the following reaction sequence



Consider **Q**, **R** and **S** as major products

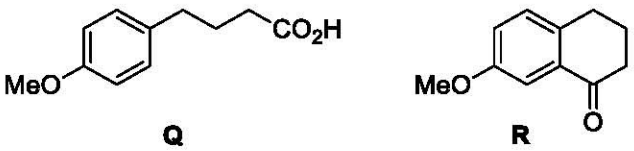
(A)



(B)



(C)



(D)



Q.8 Choose the correct option(s) from the following

(A) Natural rubber is polyisoprene containing *trans* alkene units

(B) Nylon-6 has amide linkages

(C) Teflon is prepared by heating tetrafluoroethene in presence of a persulphate catalyst at high pressure

(D) Cellulose has only α-D-glucose units that are joined by glycosidic linkages

SECTION 2 (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 If ONLY the correct numerical value is entered;
  Zero Marks : 0 In all other cases.

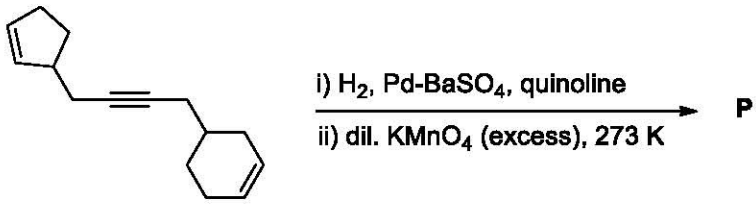
Q.9 The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc. HNO_3 to a compound with the highest oxidation state of sulphur is_____
(Given data: Molar mass of water = 18 g mol^{-1})

Q.10 Total number of *cis* N–Mn–Cl bond angles (that is, Mn–N and Mn–Cl bonds in *cis* positions) present in a molecule of *cis*-$[Mn(en)_2Cl_2]$ complex is_____ (*en* = $NH_2CH_2CH_2NH_2$)

Q.11 The decomposition reaction $2N_2O_5(g) \xrightarrow{\Delta} 2N_2O_4(g) + O_2(g)$ is started in a closed cylinder under isothermal isochoric condition at an initial pressure of 1 atm. After Y $\times$ 10^3 s, the pressure inside the cylinder is found to be 1.45 atm. If the rate constant of the reaction is 5 $\times$ 10^{-4} s^{-1}, assuming ideal gas behavior, the value of Y is_____

Q.12 The mole fraction of urea in an aqueous urea solution containing 900 g of water is 0.05. If the density of the solution is 1.2 g cm^{-3}, the molarity of urea solution is____
(Given data: Molar masses of urea and water are 60 g mol^{-1} and 18 g mol^{-1}, respectively)

Q.13 Total number of hydroxyl groups present in a molecule of the major product P is ____



Q.14 Total number of isomers, considering both structural and stereoisomers, of cyclic ethers with the molecular formula C_4H_8O is____

SECTION 3 (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO (02)** Multiple Choice Questions.
- Each List-Match set has two lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +3	If ONLY the option corresponding to the correct combination is chosen;
Zero Marks	: 0	If none of the options is chosen (i.e., the question is unanswered);
Negative Marks	: –1	In all other cases.

Answer Q.15 and Q.16 by appropriately matching the lists based on the information given in the paragraph

Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following, List-I contains some quantities for the nth orbit of the atom and List-II contains options showing how they depend on n.

List–I	List–II
(I) Radius of the nth orbit	**(P)** $\propto n^{-2}$
(II) Angular momentum of the electron in the nth orbit	**(Q)** $\propto n^{-1}$
(III) Kinetic energy of the electron in the nth orbit	**(R)** $\propto n^{0}$
(IV) Potential energy of the electron in the nth orbit	**(S)** $\propto n^{1}$
	(T) $\propto n^{2}$
	(U) $\propto n^{1/2}$

Q.15 Which of the following options has the correct combination considering List-I and List-II?

(A) (I), (P) (B) (I), (T) (C) (II), (Q) (D) (II), (R)

Q.16 Which of the following options has the correct combination considering List-I and List-II?

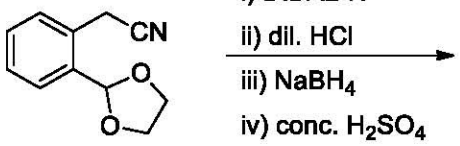
(A) (III), (P) (B) (III), (S) (C) (IV), (Q) (D) (IV), (U)

Answer Q.17 and Q.18 by appropriately matching the lists based on the information given in the paragraph

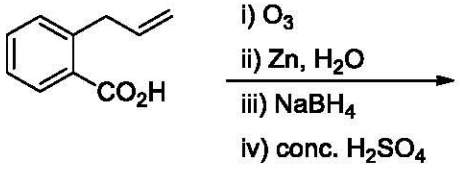
List-I includes starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of List-I.

List–I

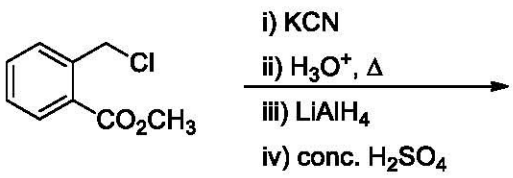
(I)



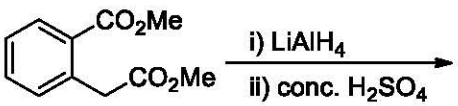
(II)



(III)

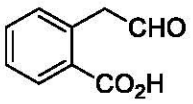


(IV)

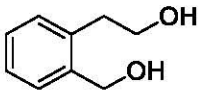


List–II

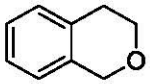
(P)



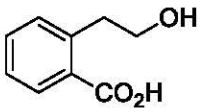
(Q)



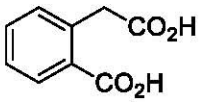
(R)



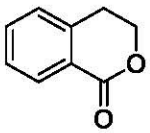
(S)



(T)



(U)



Q.17 Which of the following options has correct combination considering List-I and List-II?

(A) (I), (Q), (T), (U)

(B) (II), (P), (S), (T)

(C) (II), (P), (S), (U)

(D) (I), (S), (Q), (R)

Q.18 Which of the following options has correct combination considering List-I and List-II?

(A) (III), (S), (R)

(B) (IV), (Q), (U)

(C) (III), (T), (U)

(D) (IV), (Q), (R)